Faraz Nayeri

M.Sc. student in Process Engineering

Cologne, Germany tschembe

Profile

- Master's student in Process Engineering at TH Köln with a focus on process design, process simulation and control. Project experience along the whole chain: balances and reactor design, scale-up and cost estimation, dynamic simulation in AVEVA, CHEMCAD and MATLAB, and data-driven modelling in Python. I validate models against literature or measured data first, close the balances and document results so others can follow them.

Education

TH Köln (University of Applied Sciences), Process Engineering — Cologne, Germany
Oct 2024 – present
- Focus: process design and simulation, control engineering, scale-up and cost estimation, polymer and reaction engineering

TU Dortmund University, Studies before transferring to the master's programme at TH Köln — Dortmund, Germany
Apr 2022 – Sept 2024

Azad University, Central Tehran Branch, Chemical Engineering — Tehran, Iran
June 2014 – July 2019

Experience

TH Köln, :metabolon institute (project BiDRoGen), Intern, Hydrogen Technology — Lindlar, Germany
Aug 2026 – present
4 months
- Built a permeation model in MATLAB for iron membranes for hydrogen separation; evaluated test-rig data against literature correlations and identified the limiting transport steps.
- Built a thermodynamic stability model in Python (Cantera) to assess the oxidation risk of the membrane for different feed gases.
- Designed a lab test cell, including mass and energy balances and a mechanical check of the membrane.

TU Dortmund University, Student Assistant — Dortmund, Germany
Mar 2023 – Apr 2024
1 year 2 months
- Set up, configured and maintained a simulation environment for courses; supported students in using it.

Norara, Working Student, Chemical Production and Quality Control — Jan 2017 – Feb 2018
1 year 2 months
- Ran quality tests on raw materials and final products (FTIR, density, viscosity, pH, visual inspection).
- Checked results against product specifications, documented them for quality assurance and reported deviations.

Projects

Plantwide Control of an Ethylbenzene Plant in AVEVA Process Simulation — Mar 2026 – June 2026
Master project (10 ECTS), TH Köln
- Control concept for a plant with 2 reactors, 2 columns and 2 recycles (Buckley/Luyben): degree-of-freedom analysis (16 valves, 9 specifications) and P&IDs to DIN EN 62424.
- Full plant with custom kinetics and 16 PID controllers in AVEVA; audited the simulation files and found the cause of an oversized recycle stream.
- Independent plant model in Python (within 1.3 % of the literature). The recycle controller limits the snowball effect at +20 % throughput to +29 % instead of +91 %.

System Identification of a Slug Flow Crystallizer with Machine Learning Oct 2025 – Mar 2026

Project study TH Köln / elective project TU Dortmund

- Trained NARX networks (MATLAB, PyTorch) on about 370,000 time steps; derived the input window from the residence time and confirmed it with impulse-response analysis.
- Cut the recursive prediction error of crystal size d50 from 44 µm to 0.41 µm (R^2 = 0.999). PCA made the model 9× smaller and 4× more accurate.
- Sensitivity analysis: the cooling-jacket inlet temperature is the strongest input (+4.3 K → −14.7 µm), a basis for model-predictive control.

Electrodeposition from Deep Eutectic Solvents Oct 2025 – Jan 2026

Process Engineering Conference, TH Köln, winter term 2025/26 (Best Poster Award)

- Literature review of about 30 publications from a process engineering view; extended abstract, A0 poster and conference paper as sole author.
- Screening model in Python (mass transfer, Butler–Volmer kinetics, ohmic losses, Wagner number), validated against literature data within 4 %.
- Result: mass transfer is the limit. A DES bath needs about 19× the tank area of an aqueous bath; water and temperature together reduce this to 2.5×.

Design of a CSTR for Styrene Polymerisation Apr 2025 – Sept 2025

Module Polymer Engineering and Recycling, TH Köln

- Solved mass and energy balances with a kinetic model (incl. gel effect) in MATLAB; sized the reactor and heat exchangers; designed a two-stage cooling system; made BFD, PFD and pre-HAZOP.
- Re-checked the design in Python: model validated against literature data (within 0.22 %-points); corrected reactor volume and operating pressure (0.81 bar from vapour-liquid equilibrium).
- Simulated a loss of cooling: 43 to 65 min until 200 °C, as input for the safety review.

Scale-up and Cost Estimation of a Fluidised-Bed Reactor for Vinyl Acetate Oct 2024 – Feb 2025

Module Scale-up and Cost Estimation, TH Köln

- Mass and energy balances in CHEMCAD; scaled the reactor 200× (10 kg/h → 2 t/h) with dimensionless numbers; dynamic reactor model in MATLAB.
- AACE class 4 cost estimate (Hand factors, CEPCI): 3.0 M€ investment and about 1,940 €/t production cost.
- Confirmed the scale-up with a two-phase model (32.0 % conversion at both scales). A van Heerden analysis found an unstable operating point; PI control holds it within ±1.1 K.

Awards

Best Poster Award Jan 2026

Poster "Electrodeposition in Deep Eutectic Solvents"

27th Process Engineering Conference, TH Köln

Skills

Process engineering

Control engineering

Process simulation

Programming and data

Analytics and QA

Documentation

Languages

German
Business fluent (C1, TestDaF)

English
Good working knowledge (B2)

Persian (Farsi)
Native speaker